

Project - 4 theory

① What is statistical distribution?

Ans:- It shows how often different values or outcomes occur in a dataset.

Ex imagine measure human height. then this dist. show you how many people are short, average height or tall.

② What is g-g plot and why is it used?

Ans:- A g-g (quantile-quantile) plot is a graph that compares two sets of data to see if they follow same distribution or not.

③ difference between discrete and continuous distribution?

Ans:- discrete dist.:-

* It deal with countable values.

Ex no. of students, no. of cars

continuous dist.:-

* It deal with measurable values.

Ex height, weight, temperature

④ What is bernoulli distribution?

Ans:- It is probability dist. that models a single trial with two possible outcomes.

Ex coin toss

pass or fail

⑤ What is binomial distribution?

Ans:- It calculate the probability of a certain no. of successes in a Fixed no. of trials, where each trial has only two possible outcomes.

Ex Flip a coin 10 times and calculate probability of get exactly 7 heads.

⑥ What is log-normal distribution?

Ans:- it is probability dist. where the logarithm of the values follow normal dist.

Ex stock price, income level

⑦ What is power law distribution?

Ans:- A power law dist. means a small no. of outliers are responsible for large part of the data.

Ex

city size (few megacities, many small town)
website traffic (few popular sites, many less visited)

⑧ What is box-cox transform?

Ans:- It is statistical technique that help transform data to make it more normal and stabilize its variance.

⑨ Explain poisson distribution with an example?

Ans:- It model the no. of times an event happen in a Fixed interval (like time or space).

Ex call center receive an average of 5 calls per hour.

⑩ What is z-score probability?

Ans:- z score show how far a value from an average, measured in standard deviation.
 Ex: z-score of 2 means the value is 2 standard deviation above the average.

⑪ diff. between PDF and CDF?

Ans:- pdf:-

- * show the probability of a value occurring.
- * Think of it as the "height" of the probability curve at a specific point.

cdf:-

- * show the probability of a value being less than or equal to a specific value.
- * Think of it as the "area" under the curve" up to a specific point.